Aniruddha Upadhya K

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Education

NMAM Institute of Technology, B.E in Electronics and Communications — Dec 2021 – Present
- CGPA: 8.51/10.0

Poorna Prajna PU College, PCMCs — June 2019 – July 2021
- Percentage: 98.0%

Experience

Product Engineer Intern, Edgeverve Systems Ltd. – Bengaluru — Feb 2025 – Present
- Contributed to core codebase enhancements using *C/C++*; wrote *SQL* queries for feature development and data analysis. Worked with debugging and profiling tools including *GDB* and *Valgrind*
- Tools Used: *C, C++, GDB, Valgrind, Postgres, Oracle*

Research Intern, JP Morgan Chase & Co. – Remote — Sept 2023 - Dec 2024
- Designed and developed end to end framework that enhances security of *Federated Learning* system using *Zero-Knowledge Proofs*
- Analyzed federated learning, cryptographic methods, and aggregation strategies; compared with current approaches to evaluate framework performance
- Tools Used: *Noir, Node js, Express js, Numpy, Django, SQLite, Docker*

Patents

Mammogram Analysis and Breast Cancer Localization System and Method Thereof — Jan 2025

Shankari N, Dr. Vidya Kudva, ***Aniruddha Upadhya K***, Ashish Shankar, Ashwin Raj K R, Amarendra Kumar Singh, Mr. Shashi Kumar Shetty, Dr. Vijay Kubihal

Indian Patent Application 202541004479 A, Published Jan. 31, 2025 (Patent pending)

Projects

NixOS Config — Github
- Built a modular, declarative, and reproducible desktop setup using *NixOS* and *Home Manager*, with version-controlled configuration via *Nix Flakes*
- Configured a Wayland environment with *Hyprland, Kitty, Zsh, Neovim (via Nixvim)*, and essential dev tools
- Tools Used: *NixOS, Nix-Flake, Home-manager, Hyprland, Neovim*

Text Editor — Github
- Developed a memory-safe, Nano like text editor in *C* supporting essential navigation, editing and saving commands
- Implemented a dynamic status line displaying file info, cursor position, and real-time status messages
- Tools Used: *C, GDB, Valgrind*

Incridea'24 — Github
- Built interactive 3D pages including camera controls and character animations for Incridea'24
- Developed an *HTML5 Canvas* platformer mini-game with character motion, collision detection, sprite animations, and sound effects
- Tools Used: *Next js, Three js, React 3 Fibre, GSAP, Framer Motion, Tailwind*

Technologies

Languages: *C++, C, Javascript, Typescript, Python*

Technologies: *Postgres, GDB, Valgrind, Linux, NixOS, React js/Next js, Git, Vim/Neovim*